AutoSniper 2 - One-Page App Summary

Evidence sources in repo: README.md, DASHBOARD.py, pages/*.py, shared/data_loader.py, scripts/extract_links.py, scripts/extract_vehicle_details.py, scripts/update_bids.py, scripts/update_master.py, scripts/ai_listing_valuation.py, scripts/scheduled_jobs.py, status_app.py

What it is

AutoSniper 2 is a Streamlit toolkit for monitoring Australian vehicle auctions and maintaining live listing datasets. It combines scraping, CSV-based pipelines, and AI-assisted valuation in one codebase.

Who it is for

Primary persona: auction analysts or buyers who monitor Grays vehicle listings and need faster bid and resale decisions.

What it does

- Runs a multi-page Streamlit dashboard for live inventory, exceptions, master views, curves, and pipeline status.
- Discovers Grays listing URLs and maintains tracked active link files.
- Scrapes listing details, then writes raw, normalized, and static vehicle snapshots.
- Refreshes bid/time signals and separates records into active, sold, and referred outputs.
- Computes AI valuation outputs (risk-banded resale, recommended max bid, confidence, and verdict fields).
- Supports scheduled daily and VIC-focused refresh jobs with lock handling and internet checks.
- Loads datasets locally or syncs CSV bundles from remote URLs via AUTOSNIPER_DATA_URL settings.

How it works (architecture)

- UI layer: Streamlit app files (DASHBOARD.py + pages/) read CSV outputs and trigger scripts.
- Data layer: CSV_data/ is the primary storage (scrapers, ai, restricted, model_audit, archives, repairs).
- Ingestion and processing: extract_links -> extract_vehicle_details -> update_bids/update_master pipeline scripts.
- Enrichment: ai_listing_valuation.py uses OpenAI + repair and curve helpers, then writes ai_listing_valuations.csv.
- Orchestration and health: scheduled_jobs.py and run_nightly.py automate runs and write status/metrics.json.
- Dedicated database service: Not found in repo (CSV files are the system of record).

How to run (minimal)

- Create and activate a virtual environment in repo root: py -3 -m venv .venv, then .\.venv\Scripts\Activate.ps1.
- Install dependencies: python -m pip install --upgrade pip and python -m pip install -r requirements.txt.
- Optional config: set OPENAI_API_KEY and AUTOSNIPER_DATA_URL (or place required CSVs under CSV_data/).
- Launch UI: streamlit run DASHBOARD.py.